

James Parker

Senior Solutions Architect – AI/ML

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It is me

AI/ML Solutions Architect with 10+ years of experience designing intelligent systems, generative AI pipelines, and data platforms for large-scale enterprise and research environments. I specialize in bridging **applied machine learning and production architecture**, leading technical engagements that connect advanced research with operational excellence.

My work spans **LLM-driven data pipelines, RAG architectures, and foundation model integrations**—across AWS, GCP, and Azure. Skilled at leading POCs, conducting architecture reviews, and guiding research teams toward scalable, secure, and cost-optimized AI ecosystems. Known for distilling complex ML systems into clear, actionable designs that accelerate innovation and ensure reproducibility at scale.

Work Experience

Busy Human – Lead AI & Full-Stack Architecture

Aug 2024 – Sep 2025

Orem, UT

- Designed and implemented **enterprise-grade AI/ML architectures** supporting multimodal data, foundation model fine-tuning, and agentic orchestration.
- Partnered with **AGI/ASI research teams** to co-develop **scalable, cloud-agnostic data solutions** enabling faster model experimentation and reproducibility.
- Led **architecture validation and pilot deployments** for RAG and LLM-based systems, ensuring seamless transition from research to production.
- Built **data ingestion and transformation pipelines** using Python, LangChain, Kafka, Airflow, and Databricks for high-throughput data curation and retrieval.
- Architected **secure multi-tenant inference clusters** integrating vector databases (Pinecone, Weaviate) with Kubernetes-based deployments.
- Defined and enforced **data lineage, governance, and observability frameworks** for auditable and compliant model development.
- Collaborated with **executive stakeholders and principal engineers** to align AI infrastructure with business and compliance objectives.
- Authored **technical whitepapers and best-practice guides** on model governance, embedding optimization, and scalable RAG design.
- Implemented **zero-trust MLOps principles** leveraging IAM isolation, encryption, and artifact versioning across AWS S3, ECR, and SageMaker.
- Achieved a measurable **40% improvement in model iteration velocity** by optimizing feature-store caching and vector index orchestration.

Osmos – Full Stack & AI Engineer

Feb 2022 – May 2024

Seattle, WA

- Built **production-grade RAG pipelines** leveraging pgvector and LangChain to deliver contextual assistance directly in-app.
- Developed **embedding generation and indexing workflows** that improved retrieval precision through semantic deduplication and hybrid search.

- Implemented **evaluation harnesses** to track retrieval accuracy, hallucination rate, and latency across iterative model updates.
- Deployed **FastAPI microservices** integrating OpenAI APIs, Redis caching, and async job queues for reliable, low-latency inference.
- Designed **prompt-evaluation frameworks** and versioned datasets to ensure consistent model behavior and controllable token budgets.
- Collaborated with UX and product teams to refine AI explanations, increasing transparency and user trust in model-driven features.

CodingIT – Senior Full-Stack Developer

Mar 2021 – Dec 2021

Austin, TX

- Delivered **end-to-end product increments** across React and Node.js with automated tests and CI/CD pipelines.
- Standardized component libraries, testing utilities, and linting conventions to streamline onboarding and delivery.
- Built **authentication, billing, and analytics systems** with privacy-aware event tracking and monitoring.
- Deployed services through **GitHub Actions and Docker**, implementing environment-specific configurations and safe rollback procedures.
- Partnered with clients using RFC-based planning and demo-driven delivery to align scope, budget, and outcomes.

CraftedQ – Full Stack Engineer

Nov 2015 – Feb 2021

Houston, TX

- Migrated legacy systems to **React + TypeScript** and modular Node.js microservices.
- Introduced **automated testing and CI/CD pipelines** (Jenkins → GitHub Actions) that cut regression incidents significantly.
- Led performance remediation across SQL, caching, and front-end assets, improving load times for high-traffic pages.
- Collaborated on **OpenAPI specifications** and schema governance to ensure backward-compatible integrations.

GoMage – Software Engineer

Jul 2011 – Aug 2015

San Antonio, TX

- Developed **e-commerce modules** with secure payment and shipping integrations, optimizing reliability and audit trails.
- Implemented **build automation** (Grunt → Gulp) to standardize pipelines and accelerate front-end feedback cycles.
- Optimized UI performance and accessibility, reducing bounce rates and improving conversion stability during high-volume campaigns.

Education

University of North Texas – BS, Computer Science

Skills

- ✓ **AI/ML Stack:** PyTorch, TensorFlow, LangChain, LangGraph, LlamaIndex, MLflow, Airflow, Databricks, Spark, Kafka, RAG, Vector Databases (Pinecone, FAISS, Weaviate), Feature Stores, Data Lakes
- ✓ **Programming Languages:** Python, TypeScript, Java, SQL, PyTorch, TensorFlow, FastAPI, LangChain, Flask
- ✓ **Front-End (React):** React 18, Next.js 13/14 (App Router, SSR/ISR), React Query/TanStack Query, Hooks & Context, Zustand/Redux, React Hook Form, **Zod** schemas, Tailwind CSS, CSS Modules, Sass, Accessibility (WCAG 2.2/ARIA), Code-splitting & Suspense
- ✓ **Back-End (Node):** Node.js 14–20, Express and Nest-style patterns, Fastify, RESTful APIs, GraphQL (Apollo), WebSockets/SSE, Webhooks (signed/HMAC), Background jobs & queues, Schedulers/Cron, Caching & rate limiting with **Redis**, Idempotency keys, OpenAPI/Swagger specs
- ✓ **APIs & Integrations:** OAuth 2.1/OIDC, JWT & session flows, RBAC/ABAC, Payments & billing, Email/SMS providers, Third-party webhooks, File uploads & object storage (S3-compatible), Audit trails & event logs
- ✓ **Data & Storage:** PostgreSQL (schema design, migrations, indexing, performance tuning), MySQL, MongoDB, Redis (cache, queues, counters), Query optimization & EXPLAIN, Safe migrations & backfills
- ✓ **Observability & Quality: OpenTelemetry** traces/metrics/logs, health checks & RED/USE dashboards, SLOs & alerts, incident runbooks, regression triage, Testing: **Jest**, **Vitest**, Testing Library, **Cypress**, **Playwright**, API tests (Supertest), **contract tests (Pact)**
- ✓ **Security & Reliability:** Input validation (**Zod**), CSRF/XSS/SSRF mitigations, headers & CSP, secrets management, dependency hygiene, abuse prevention & throttling, blue/green & canary releases, safe rollbacks
- ✓ **DevEx & Delivery:** GitHub Actions CI/CD, Docker images & multi-stage builds, environment configuration, feature flags, RFCs/ADRs, code review guidelines, story mapping & demo-driven development, change logs & release notes

Hobbies/Interests

- ❖ Distance running and strength training with measurable goals; apply sports-science principles to sustainable performance—similar to how I approach reliability work.
- ❖ Woodworking and DIY projects with a focus on safety, clean joinery, and careful planning; habits that translate to disciplined engineering practices.
- ❖ Photography and urban sketching that sharpen composition, hierarchy, and whitespace—useful instincts when crafting accessible, legible UIs.
- ❖ Volunteering at local STEM events; mentor students building their first web apps with an emphasis on curiosity and respectful collaboration.